

SAT Math Test IV

1

★ Mark for Review

ABC

In the system of equations below, c is a constant. If the system has no solution, what is the value of c ?

$$y = \frac{1}{2}x + 8$$

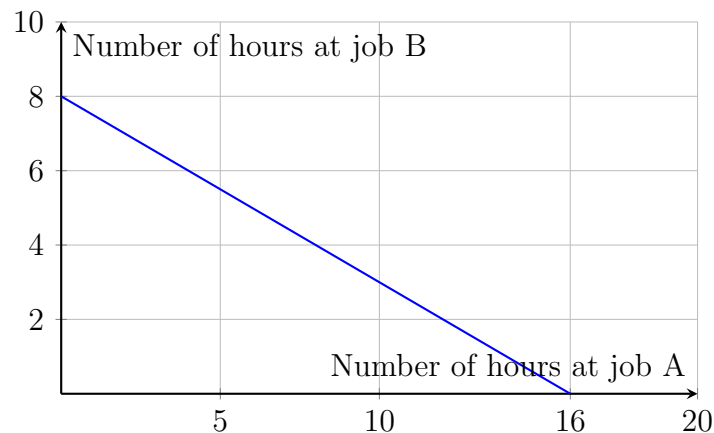
$$y = cx + 10$$

2

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ABC

To earn money for college, Avery works two part-time jobs: A and B. She earns \$10 per hour working at job A and \$20 per hour working at job B. In one week, Avery earned a total of s dollars for working at the two part-time jobs. The graph below represents all possible combinations of numbers of hours Avery could have worked at the two jobs to earn s dollars. What is the value of s ?



A. 128

B. 160

C. 200

D. 320

3

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ABC

For each real number r , which of the following points lies on the graph of each equation in the xy -plane for the given system?

$$\begin{aligned}2x + 3y &= 7 \\ 10x + 15y &= 35\end{aligned}$$

A. $\left(\frac{7}{5} + r, -\frac{7}{5} + 35\right)$

B. $\left(-\frac{3r}{2} + \frac{7}{2}, r\right)$

C. $\left(r, \frac{2r}{3} + \frac{7}{3}\right)$

D. $\left(r, -\frac{3r}{2} + \frac{7}{2}\right)$

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ABC

The triangle inequality theorem states that the sum of any two sides of a triangle must be greater than the length of the third side. If a triangle has side lengths of 6 and 12, which inequality represents the possible lengths, x , of the third side of the triangle?

A. $x < 18$

B. $x > 18$

C. $6 < x < 18$

D. $x < 6$ or $x > 18$

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ABC

The expression $\frac{x^{-2}y^{\frac{1}{2}}}{x^{\frac{1}{3}}y^{-1}}$, where $x > 1$ and $y > 1$, is equivalent to which of the following?

A. $\frac{\sqrt{y}}{\sqrt[3]{x^2}}$

B. $\frac{y\sqrt{y}}{\sqrt[3]{x^2}}$

C. $\frac{y\sqrt{y}}{x\sqrt{x}}$

D. $\frac{y\sqrt{y}}{x^2\sqrt[3]{x}}$

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ABC

Which expression is equivalent to $\frac{y+12}{x-8} + \frac{y(x-8)}{x^2y-8xy}$?

A. $\frac{xy+y+4}{x^2y-16x+64xy}$

B. $\frac{xy+9+12}{x^2y-8xy+x-8}$

C. $\frac{xy^2+13xy-8y}{x^2y-8xy}$

D. $\frac{xy^2+13xy-8y}{x^2y-16x^2y+64xy}$

7

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ABC

Solve for x : $\sqrt{2x+6} + 4 = x + 3$ What is the solution set of the equation above?

A. $\{-1\}$

B. $\{5\}$

C. $\{-1, 5\}$

D. $\{0, -1, 5\}$

The formula

$$D = T - \frac{9}{25}(100 - H)$$

approximates the dew point D in Fahrenheit, given temperature T in Fahrenheit, and relative humidity $H\%$, where $H > 50$. Which of the following expresses H in terms of D and T ?

A. $H = \frac{25}{9}(D - T) + 100$

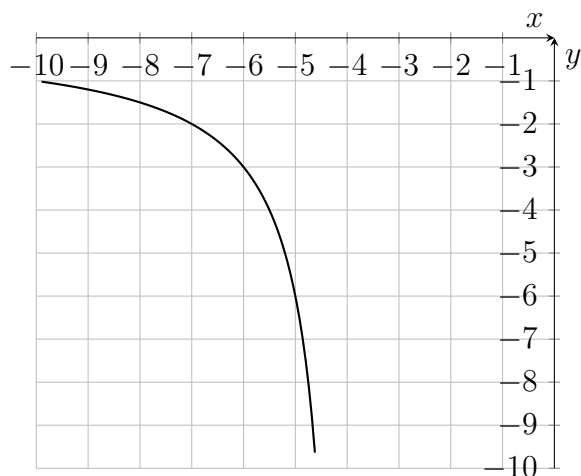
B. $H = \frac{25}{9}(D - T) - 100$

C. $H = \frac{25}{9}(D + T) + 100$

D. $H = \frac{25}{9}(D + T) - 100$

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ABC

The rational function f is defined by an equation in the form $f(x) = \frac{a}{x+b}$, where a and b are constants. The partial graph of $y = f(x)$ is shown. If $g(x) = f(x+4)$, which equation could define function g ?

A. $g(x) = \frac{6}{x}$

B. $g(x) = \frac{6}{x+4}$

C. $g(x) = \frac{6}{x+8}$

D. $g(x) = \frac{6(x-4)}{x+4}$

10

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ABC

In the equation $x^2 - 34x + c = 0$, c is a constant. The equation has no real solutions if $c > n$. What is the least possible value of n ?

11

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ABC

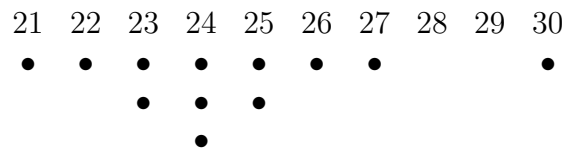
The number w is 110% greater than the number z . The number z is 55% less than 50. What is the value of w ?

12

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ABC

The dot plot below shows the gas mileages, in miles per gallon, of 12 different cars.



If the dot representing the car with the greatest gas mileage is removed from the dot plot, which of the following measures of the data set will decrease?

- I. The mean
- II. The median
- III. The standard deviation

A. I only**B.** I and III only**C.** II and III only**D.** I, II, and III

13

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ABC

In a psychological study, 600 college gymnasts were randomly selected and put in high-stakes competitions to determine whether they performed better under pressure. Based on the results of this study, it is estimated that 65% of all college gymnasts perform better under pressure, with an associated margin of error of 4%.

Which of the following inferences can appropriately be drawn based on the given margin of error?

A. The percentage of all college gymnasts who perform better under pressure is between 65% and 69%.

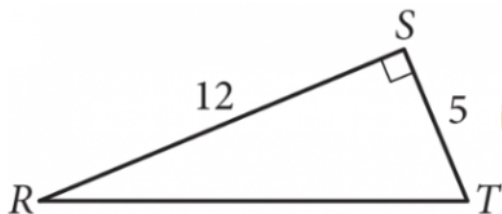
B. Approximately 4% of the college gymnasts in the study who performed better under pressure do not actually perform better under pressure.

C. It is unlikely that the percentage of all college gymnasts who perform better under pressure is 60%.

D. It is not possible that the percentage of all college gymnasts who perform better under pressure is 70%.

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ABC

In triangle RST above, point W (not shown) lies on $\overline{RT}$. What is the value of $\cos(\angle RSW) - \sin(\angle WST)$?

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ABC

A regular polygon has an interior angle that is 4 times the exterior angle plus 60 degrees. What is the number of sides of the polygon?

A. 9

B. 10

C. 12

D. 15

16

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ABC

A circle in the xy -plane has equation

$$(x - 2.5)^2 + (y - 3.5)^2 = 225.$$

Line p is tangent to the circle at point $(-6.5, C)$, where C is a constant. Which of the following could be the slope of line p ?

I. $-\frac{3}{4}$ II. $\frac{3}{4}$ III. $4/3$

A. I only

B. I and II only

C. I and III only

D. I, II, and III

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ABC

The function f is defined by $f(x) = a^x - b$, where a and b are constants. In the xy -plane, the graph of $y = f(x)$ passes through the points $(c, 7)$ and $(2c, 117)$, where c is a constant. Which of the following could be the value of b ?

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ABC

The given equation, where t is a positive constant, defines a circle in the xy -plane. The radius of this circle is $\sqrt{41}$. What is the value of t ?

$$2x^2 + 2y^2 + tx + ty - 78 = 0$$

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ABC

The table summarizes the distribution of heights, in inches (“in”), for 300 participants in a study. If a participant that is at most 71 inches tall is chosen at random, what is the probability that the participant was selected from Group 2?

	48–59 in.	60–71 in.	72–83 in.	Total
Group 1	20	60	20	100
Group 2	10	75	15	100
Group 3	30	45	25	100
Total	60	180	60	300

A. $\frac{1}{4}$

B. $\frac{17}{48}$

C. $\frac{17}{60}$

D. $\frac{17}{20}$

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ABC

The function f is defined as $f(x) = x^2 + ax + b$, where a and b are positive integers. If the minimum value of $f(x)$ is 7, which is a possible value of b ?

A. 6

B. 7

C. 8

D. 9

21

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ABC

The average annual energy cost for a certain home is \$4,334. The homeowner plans to spend \$25,000 to install a geothermal heating system. The homeowner estimates that the average annual energy cost will then be \$2,712. Which of the following inequalities can be solved to find t , the number of years after installation at which the total amount of energy cost savings will exceed the installation cost?

A. $25,000 > (4,334 - 2,712)t$ B. $25,000 < (4,334 - 2,712)t$ C. $25,000 - 4,334 - 2,712t$ D. $25000 > \frac{4332}{2712}t$

22

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ABC

Given the quadratic function

$$y = x^2 - 2ax + a \quad (a \neq 0)$$

passes through two points

$$A\left(\frac{a}{2}, y_1\right), \quad B(3a, y_2),$$

which of the following statements is correct?

A. There exists a real number a such that $y_1 > a$.

B. For all real numbers a , $y_1 > a$ is always true.

C. There exists a real number a such that $y_2 < 0$.

D. For all real numbers a , $y_2 < 0$ is always true.